Root test - 5.6

Root test: Let $\sum_{n=1}^{\infty} a_n$ be a series with
$\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$.

i. If $0 \le L < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely.

ii. If $L > 1$ or $L = \infty$, then $\sum_{n=1}^{\infty} a_n$ diverges.

iii. If $L = 1$, the test is inconclusive.

$$\sqrt[n]{|a_n|} \approx L \iff |a_n| \approx L^n$$

Eventually, $\sum a_n$ looks like $\dots + L^n + L^{n+1} + L^{n+2} + \dots$

<u>Example</u>: ATCO $\sum_{n=1}^{\infty}\left(\frac{2n^2+1}{3n^2+2n+1}\right)^n$.

Seeing <u>everything</u> raised to the n^{th} power suggests the root test.

$$\lim_{n\to\infty}\sqrt[n]{|a_n|} = \lim_{n\to\infty}\sqrt[n]{\left|\left(\frac{2n^2+1}{3n^2+2n+1}\right)^n\right|}$$

$= \lim_{n\to\infty}\left|\frac{2n^2+1}{3n^2+2n+1}\right| = \left|\frac{2}{3}\right| < 1$, so by the root test, $\sum_{n=1}^{\infty}\left(\frac{2n^2+1}{3n^2+2n+1}\right)^n$ converges absolutely.

<u>Example</u>: ATCO $\sum_{n=1}^{\infty} \frac{1}{(\ln(n))^n}$

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\frac{1}{(\ln(n))^n}\right|} = \lim_{n\to\infty} \left|\frac{1}{\ln(n)}\right| = 0$$

and $0 < 1$, so $\sum_{n=1}^{\infty} \frac{1}{(\ln(n))^n}$ converges absolutely by the root test.

Example: ATCO $\displaystyle\sum_{n=1}^{\infty} \frac{(\ln(n))^n}{n}$:

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \left| \frac{\ln(n)}{\sqrt[n]{n}} \right| = \lim_{n\to\infty} \left| \frac{\overbrace{\ln(n)}^{\to\infty}}{\underbrace{n^{\frac{1}{n}}}_{\to 1}} \right| = \infty$$

So by the root test, $\displaystyle\sum_{n=1}^{\infty} \frac{(\ln(n))^n}{n}$ diverges.